KU LEUVEN

FACULTY OF ENGINEERING TECHNOLOGY

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Miniature and customer friend solution for tracking and locating insects

Subtitle (optional)

Lars DEPREZ

Supervisor(s): Prof. dr. ing. G Callebaut
ing. J Van Moer

Co-supervisor(s): dr. ing. J Van Mulders

Master's Thesis submitted to obtain the degree of Master of Science in ghenteng specialisationprogramme Engineering Technology

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Summary

In recent years, the invasive Asian hornet (*Vespa velutina*) has appeared across Europe, posing a significant threat to local wildlife and apiaries. Multiple methods for containing this threat have been invented: including visual tracking, triangulation, harmonic radar and radio telemetry.

This thesis investigates radio telemetry to track Asian hornets. The primary focus is to design a cost-effective, rechargeable, sub-200 mg transmitter that is practical and user-friendly for non-technical operators. The user must be able to attach and activate the transmitter in a simple manner and track the hornet with a portable receiver. This thesis aims to improve upon existing wildlife trackers by being purpose-built specifically for the tracking and eradication of Asian hornets.

The transmitter must minimize energy consumption and component count. This is achieved using a simple, single-stage transmitter designed around a Pierce oscillator combined with a pulsing circuit to conserve energy.

Weight minimization is a critical aspect of this research, offering key practical advantages: reduced tag weight allows a larger battery to be integrated to extend operational lifespan or increase RF output power, while enabling a higher proportion of hornets to carry the payload over greater distances.

The primary goals of this thesis are first to design, simulate, and fabricate an initial transmitter tag, which will be evaluated and refined into an optimized second design iteration. Once a functional tag is realized, a complementary receiver will be designed and tested, leading to field deployment once the tag is small and light enough.

Keywords: Vespa velutina, radio telemetry, Pierce oscillator

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Miniature and customer friendly solution for tracking and locating insects

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1 introduction

1.1 Asain hornet

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1.2 Problem statement

1.3 Research Questions

1.3.1 Conclusions

2 Literature review

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3 Design and validation

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[1] B. Naef-Daenzer, "A New Transmitter for Small Animals and Enhanced Methods of Home-Range Analysis", *The Journal of Wildlife Management*, vol. 57, no. 4, pp. 680–689, 1993, issn: 0022-541X. doi: 10.2307/3809066 JSTOR: 3809066. Accessed: Oct. 13, 2026. [Online]. Available: https://www.jstor.org/stable/3809066

Appendix A

The crystal oscillator

In this appendix, the theory of the single-stage transmitter is explained. This is a simple transmitter with one transistor and a crystal. The explanation is divided into multiple parts, starting with the theory of the crystal, followed by the Pierce circuit. Then, we calculate the minimum transconductance needed by the transistor, and finally end with the design of the LC tank circuit and antenna.

Figure A.1 displays the single-stage transmitter circuit, which will be analyzed step by step in the following sections.

This circuit works based on two 180° phase shifts of the signal: once by the transistor and once by the crystal. The condition when this phase shift occurs is described below.

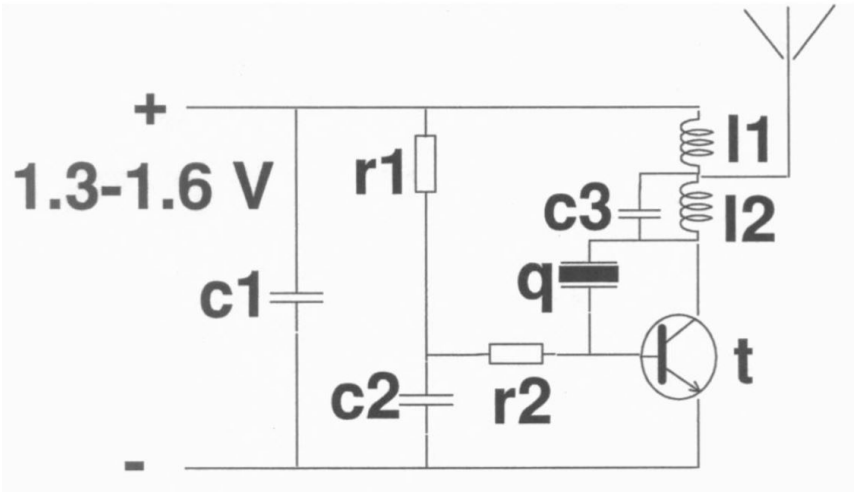


Figure A.1: Schematic of the single-stage Pierce oscillator circuit (taken from [1]).

A.1 Quartz Crystal

The quartz crystal sits centrally in the schematic and creates the stable RF signal of the circuit. To achieve the required VHF operating frequency while keeping the crystal physically robust, the crystal is operated in its 3rd overtone mode (approximately three times its fundamental frequency). To analyze its electrical behavior and frequency response, the crystal is represented by the Butterworth-Van Dyke (BVD) equivalent circuit model shown in Figure A.2.

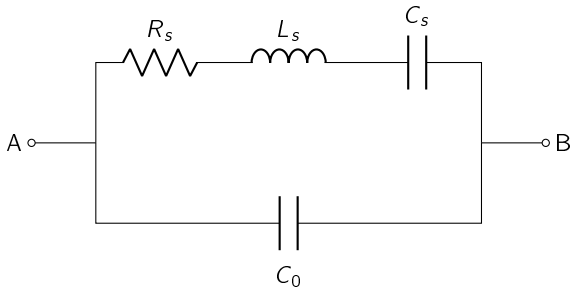


Figure A.2: Butterworth-Van Dyke (BVD) equivalent circuit model of a quartz crystal showing the series motional branch (R_s, L_s, C_s) in parallel with the static shunt capacitance (C_0) (own work).

The frequency response is derived by combining the series and parallel branches together and determining the resonance conditions.

$$Z_s(j\omega) = j\omega L_s + \frac{1}{j\omega C_s} = \frac{1-\omega^2 L_s C_s}{j\omega C_s}, \quad Z_0(j\omega) = \frac{1}{j\omega C_0} \tag{A.1}$$

Combining both equations in parallel yields:

$$\begin{aligned} Z(j\omega) &= Z_s(j\omega) \parallel Z_0(j\omega) \\ &= \frac{\left(\frac{1-\omega^2 L_s C_s}{j\omega C_s}\right) \cdot \left(\frac{1}{j\omega C_0}\right)}{\left(\frac{1-\omega^2 L_s C_s}{j\omega C_s}\right) + \frac{1}{j\omega C_0}} \\ &= \frac{1-\omega^2 L_s C_s}{j\omega\left[(C_s + C_0) - \omega^2 L_s C_s C_0\right]} \\ &= \frac{1}{j\omega C_0} \cdot \frac{\frac{1}{L_s C_s} - \omega^2}{\frac{1}{L_s}\left(\frac{1}{C_s} + \frac{1}{C_0}\right) - \omega^2} \end{aligned} \tag{A.2}$$

Finding the zeros and poles of $Z(j\omega)$ shows the resonant frequencies of the quartz crystal:

1. **Series Resonant Frequency (f_s):** Occurs when $Z(j\omega) = 0$:

$$f_s = \frac{1}{2\pi\sqrt{L_s C_s}} \tag{A.3}$$

2. **Parallel Resonant Frequency (f_p):** Occurs when $Z(j\omega) \to \infty$. Factoring out C_s directly relates f_p to f_s:

$$f_p = \frac{1}{2\pi}\sqrt{\frac{1}{L_s C_s}\left(1 + \frac{C_s}{C_0}\right)} = f_s\sqrt{1 + \frac{C_s}{C_0}} \tag{A.4}$$

Since $C_0 \gg C_s$, the ratio $\frac{C_s}{C_0} \ll 1$. Applying a first-order Taylor series approximation ($(1 + x)^{1/2} \approx 1 + \frac{x}{2}$) provides a simplified expression for the parallel resonant frequency:

$$f_p \approx f_s\left(1 + \frac{C_s}{2C_0}\right) \tag{A.5}$$

For $f < f_s$ and $f > f_p$, the circuit behaves like a capacitor, whereas for $f_s < f < f_p$, it behaves like an inductor. These properties will be important when defining the oscillation conditions in the Pierce circuit.

A.2 Pierce oscillator

The Pierce oscillator works on the basis that a quartz crystal and two capacitors will induce a 180° phase shift onto a signal. Inverting that signal then results in an oscillation following the Barkhausen criteria, meaning the total phase shift must be $k \cdot 360°$ ($k \in \mathbb{N}$) and the amplification should be ≥ 1.

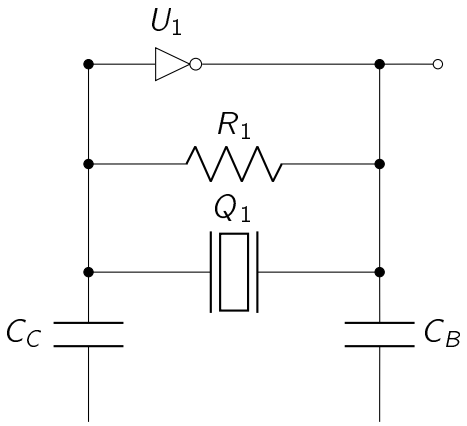


Figure A.3: A Pierce oscillator circuit.

In the following steps, C_b and C_c represent the total node capacitances from base to ground and collector to ground, respectively. The feedback resistor and inverter are removed in this analysis, as the transistor functions as the inverting element.

We start by modeling the collector-to-base voltage ratio using a voltage divider formed by the crystal reactance X_q and the base capacitance C_b:

$$Z_b(j\omega) = \frac{1}{j\omega C_b}, \quad Z_q(j\omega) = jX_q \tag{A.6}$$

Applying the voltage divider rule yields the base voltage V_b relative to the collector voltage V_c:

$$\begin{aligned}\frac{V_b}{V_c} &= \frac{Z_b(j\omega)}{Z_q(j\omega) + Z_b(j\omega)} \\ &= \frac{\frac{1}{j\omega C_b}}{jX_q + \frac{1}{j\omega C_b}} \\ &= \frac{1}{1 - \omega C_b X_q}\end{aligned} \tag{A.7}$$

To achieve the required 180° phase shift, the transfer ratio must be negative ($\frac{V_b}{V_c} < 0$). This requires the denominator $1 - \omega C_b X_q$ to be negative. Since ω and C_b are strictly positive, X_q must be positive ($X_q > 0$), confirming that the crystal must operate in its inductive reactance region ($f_s < f < f_p$).

An important effect of the node capacitances C_b and C_c is that they alter the resonant frequency of the quartz crystal, which must be taken into account. Looking from the crystal terminals, C_b and C_c are in series with each other and in parallel with C_0. The total external load capacitance C_L experienced by the crystal is defined as:

$$C_L = \frac{C_b C_c}{C_b + C_c} \tag{A.8}$$

In addition to the static capacitance C_0 and the external load capacitance C_L, the internal transistor feedback capacitance C_μ (between collector and base) also sits directly in parallel across the crystal terminals. Combining these contributions yields the total effective parallel capacitance C_T:

$$C_T = C_0 + C_L + C_\mu = C_0 + \frac{C_b C_c}{C_b + C_c} + C_\mu \tag{A.9}$$

Consequently, the loaded parallel resonant frequency f_p expressed in terms of C_T becomes:

$$f_p = f_s \sqrt{1 + \frac{C_s}{C_T}} \tag{A.10}$$

Applying the first-order Taylor approximation ($(1 + x)^{1/2} \approx 1 + \frac{x}{2}$) yields the simplified expression for the loaded frequency:

$$f_p \approx f_s \left(1 + \frac{C_s}{2C_T}\right) \tag{A.11}$$

A.3 Transconductance calculation

In this section, the critical transconductance ($g_{m,\text{crit}}$) of the transistor is derived to determine the threshold condition for oscillation startup. This is done by deriving the negative resistance generated by the BJT and adjusting for losses inside the quartz crystal.

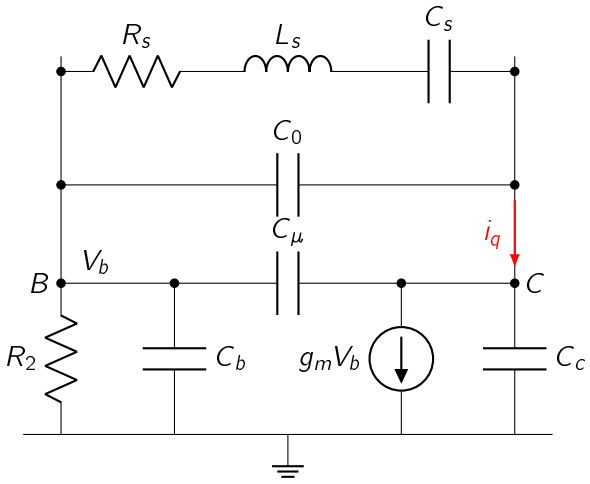


Figure A.4: Small-signal equivalent circuit of the BJT Pierce oscillator with current i_q flowing from the crystal feedback network into the collector node.

Node B Analysis

The bias resistor R_2 can be neglected assuming it is sufficiently large.

$$V_b = -\frac{i_q}{j\omega C_b} \tag{A.12}$$

Node C Analysis (KCL)

Applying KCL at Node C, the sum of currents entering the node equals the sum of currents leaving to ground.

$$i_q = g_m V_b + j\omega C_c V_c \tag{A.13}$$

Solving for V_c directly from the Node C:

$$V_c = \frac{i_q - g_m V_b}{j\omega C_c} \tag{A.14}$$

Substituting V_b into this expression gives:

$$\begin{aligned} V_c &= \frac{i_q - g_m\left(-\frac{i_q}{j\omega C_b}\right)}{j\omega C_c} \\ &= \frac{i_q}{j\omega C_c} - \frac{g_m i_q}{\omega^2 C_b C_c} \end{aligned} \tag{A.15}$$

Combining these two nodes with i_q results into the impedance seen by the quartz crystal.

$$\begin{aligned} V_c - V_b &= \left(\frac{i_q}{j\omega C_c} - \frac{g_m i_q}{\omega^2 C_b C_c}\right) - \left(-\frac{i_q}{j\omega C_b}\right) \\ &= i_q\left(\frac{1}{j\omega C_b} + \frac{1}{j\omega C_c} - \frac{g_m}{\omega^2 C_b C_c}\right) \\ &= i_q\left(-\frac{g_m}{\omega^2 C_b C_c} - j\frac{C_b + C_c}{\omega C_b C_c}\right) \end{aligned} \tag{A.16}$$

Dividing by i_q defines the active transistor input impedance Z_t seen by the crystal:

$$Z_t = \frac{V_c - V_b}{i_q} = -\frac{g_m}{\omega^2 C_b C_c} - \frac{j}{\omega C_L} \tag{A.17}$$

where $C_L = \frac{C_b C_c}{C_b + C_c}$ is the equivalent series load capacitance. The negative real part $R_a = -\frac{g_m}{\omega^2 C_b C_c}$ provides the active power necessary to overcome system losses and initiate oscillation.

Power Balance and Critical Transconductance

To determine the startup threshold condition, the crystal current i is modeled as dividing between two parallel branches across the terminal voltage v:

- **Branch A (Active Circuit):** Impedance $Z_t = R_a - \frac{j}{\omega C_L}$
- **Branch P (Parasitics):** Combined capacitance $C_p = C_0 + C_\mu$, with impedance $Z_p = \frac{1}{j\omega C_p}$

Applying Kirchhoff's Current Law (KCL) at the node and expressing the currents using branch admittances ($Y = \frac{1}{Z}$):

$$i = i_a + i_p \tag{A.18}$$

$$v = i_a Z_t = i_p Z_p \implies i_a = vY_t, \quad i_p = vY_p \tag{A.19}$$

Summing the branch currents allows v to be expressed in terms of the total current i:

$$i = v(Y_t + Y_p) \implies v = \frac{i}{Y_t + Y_p} \tag{A.20}$$

Substituting v back into the active branch current $i_a = vY_t$ gives the exact current divider equation:

$$i_a = i\frac{Y_t}{Y_t + Y_p} \tag{A.21}$$

The branch admittances are defined as $Y_p = j\omega C_p$ and $Y_t = \frac{1}{R_a - j/(\omega C_L)}$. Assuming $|R_a| \ll \frac{1}{\omega C_L}$, the active impedance is dominated by its capacitive term ($Z_t \approx -\frac{j}{\omega C_L}$), simplifying the active admittance to $Y_t \approx j\omega C_L$.

Substituting Y_t and Y_p back into the current divider yields:

$$\begin{aligned} i_a &= i\frac{j\omega C_L}{j\omega C_L + j\omega C_p} \\ &= i\frac{C_L}{C_L + C_p} \\ &= i\frac{C_L}{C_L + C_0 + C_\mu} = i\frac{C_L}{C_T} \end{aligned} \tag{A.22}$$

where $C_T = C_L + C_0 + C_\mu$ represents the total parallel capacitance.

Real power is exchanged solely within the active resistance R_a (driven by i_a) and the crystal's series loss resistance R_s (driven by i):

$$P_{\text{supplied}} = |R_a| i_a^2 = |R_a| \left(\frac{C_L}{C_T}\right)^2 i^2 \tag{A.23}$$

$$P_{\text{loss}} = R_s i^2 \tag{A.24}$$

Equating $P_{\text{supplied}} = P_{\text{loss}}$ and canceling i^2:

$$R_s = |R_a| \left(\frac{C_L}{C_T}\right)^2 \tag{A.25}$$

Substituting $|R_a| = \frac{g_m}{\omega^2 C_b C_c}$ yields the final critical transconductance $g_{m,\text{crit}}$ required for oscillation startup:

$$g_{m,\text{crit}} = \omega^2 C_b C_c R_s \left(\frac{C_T}{C_L}\right)^2 \tag{A.26}$$

The resulting $g_{m,\text{crit}}$ is used to calculate the required base bias current. To ensure a reliable startup, it is recommended to maintain a design margin of $g_m = (3 \text{ to } 5) \times g_{m,\text{crit}}$.

The DC collector bias current I_c is determined from the transconductance g_m and the thermal voltage V_T:

$$I_c = g_m V_T \tag{A.27}$$

where $V_T = \frac{k_B T}{q} \approx 25.85$ mV at room temperature (300 K).
The corresponding base bias current I_b is obtained using the transistor's common-emitter current gain β:

$$I_b = \frac{I_c}{\beta} = \frac{g_m V_T}{\beta} \tag{A.28}$$

Keep in mind that V_T and β fluctuate with temperature and operating conditions, so extreme values should be taken into account when creating a practical design.

A.4 Collector arm

The Pierce oscillator requires a capacitive reactance on both sides of the quartz crystal. On the collector side, this capacitance is named C_c, but in reality, it consists of a network comprising two inductors, a capacitor, and an antenna. For the oscillator to operate, this network must have a net capacitive reactance at the oscillation frequency.

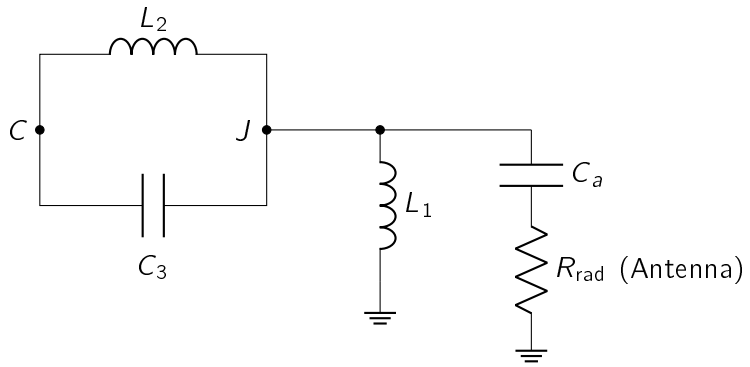


Figure A.5: Equivalent circuit representation of the collector network connected to the parallel $L_2//C_3$ tank, L_1 matching inductor, and the antenna load model (C_a, R_{rad}).

The lower bound of the capacitive frequency range for the collector arm is defined by $\omega_{\text{tank}} = \frac{1}{\sqrt{L_2 C_3}}$, where $L_1 \parallel C_a$ is simplified to $L_1' = \frac{L_1}{1-\omega^2 L_1 C_a}$.
The upper bound is obtained by requiring a net capacitive reactance:

$$\omega L_1' + \frac{\omega L_2}{1 - \omega^2 L_2 C_3} < 0 \tag{A.29}$$

Dividing by $\omega > 0$ and multiplying by $(1 - \omega^2 L_2 C_3) < 0$ (which reverses the inequality sign since $\omega > \omega_{\text{tank}}$) yields:

$$\begin{aligned} L_2 &> -L_1' \left(1 - \omega^2 L_2 C_3\right) \\ L_1' + L_2 &> \omega^2 L_1' L_2 C_3 \\ \omega^2 &< \frac{1}{L_2 C_3}\left(1 + \frac{L_2}{L_1'}\right) \end{aligned} \tag{A.30}$$

Substituting $\omega_{\text{tank}}^2 = \frac{1}{L_2 C_3}$ defines the valid frequency range for capacitive collector behavior:

$$\omega_{\text{tank}} < \omega_{\text{osc}} < \omega_{\text{tank}} \sqrt{1 + \frac{L_2}{L_1'}} \tag{A.31}$$

The tank circuit must be designed such that the oscillation frequency falls within these frequency bounds.

A.5 The antenna

A.6 Design formula summary

For convenience during the practical design phase, all critical equations required to calculate the components and operating conditions of the single-stage Pierce oscillator are summarized below.

Crystal and Load Capacitance

$$f_s = \frac{1}{2\pi\sqrt{L_s C_s}} \quad \text{(Intrinsic series resonant frequency)} \tag{A.32}$$

$$C_L = \frac{C_b C_c}{C_b + C_c} \quad \text{(External load capacitance)} \tag{A.33}$$

$$C_T = C_0 + C_L + C_\mu \quad \text{(Total effective parallel capacitance)} \tag{A.34}$$

$$f_L \approx f_s \left(1 + \frac{C_s}{2C_T}\right) \quad \text{(Loaded operating frequency)} \tag{A.35}$$

Transistor Startup and Biasing

$$g_{m,\text{crit}} = \omega^2 C_b C_c R_s \left(\frac{C_T}{C_L}\right)^2 \quad \text{(Minimum critical transconductance)} \tag{A.36}$$

$$g_m = (3 \text{ to } 5) \times g_{m,\text{crit}} \quad \text{(Target transconductance with safety margin)} \tag{A.37}$$

$$I_c = g_m V_T \quad \text{(Required DC collector bias current)} \tag{A.38}$$

$$I_b = \frac{I_c}{\beta} \quad \text{(Required DC base bias current)} \tag{A.39}$$

Collector Network and Antenna Matching

$$L_1' = \frac{L_1}{1 - \omega^2 L_1 C_a} \quad \text{(Equivalent matching inductance, assuming } L_1' > 0\text{)} \quad \text{(A.40)}$$

$$\omega_{\text{tank}} = \frac{1}{\sqrt{L_2 C_3}} \quad \text{(Lower boundary of capacitive region)} \quad \text{(A.41)}$$

$$\omega_{\text{osc}} < \omega_{\text{tank}} \sqrt{1 + \frac{L_2}{L_1'}} \quad \text{(Upper boundary of capacitive region)} \quad \text{(A.42)}$$

$$L_1 \approx \frac{1}{\omega_{\text{osc}}^2 C_a} \quad \text{(Optimal matching inductor for antenna resonance)} \quad \text{(A.43)}$$

Appendix B

Scientific article describing this thesis

The thesis should also contain a short scientific article. If you write your thesis in Dutch, you must write the article in English, and vice versa. We advise you to employ the IEEE Manuscript Templates for Conference Proceedings (`https://www.overleaf.com/latex/templates/ieee-conference-template/grfzhhncsfqn`). Compile the article in another project and include the generated pdf file as shown below:

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C. Equations

Number equations consecutively. To make your equations more compact, you may use the solidus (/), the exp function, or appropriate exponents. Italicize Roman symbols for quantities and variables, but not Greek symbols. Use a long dash rather than a hyphen for a minus sign. Punctuate equations with commas or periods when they are part of a sentence, as in:

$$a + b = \gamma \tag{1}$$

Be sure that the symbols in your equation have been defined before or immediately following the equation. Use “(1)”, not

Appendix C

Poster

Appendix D

GenAI code of conduct

Placeholder

GenAI code of conduct

Download the most recent KU Leuven GenAI code of conduct, fill it in, export the completed document to PDF, and manually include that PDF here.

The files `chapters/GenAI_Code of conduct_AJ 2526_*` are provided as a starting point only. Always check the KU Leuven website for the most recent version before submitting your thesis.

Replace this placeholder with, for example:

```
\includepdf [pages=-]{chapters/code-of-conduct.pdf}
```